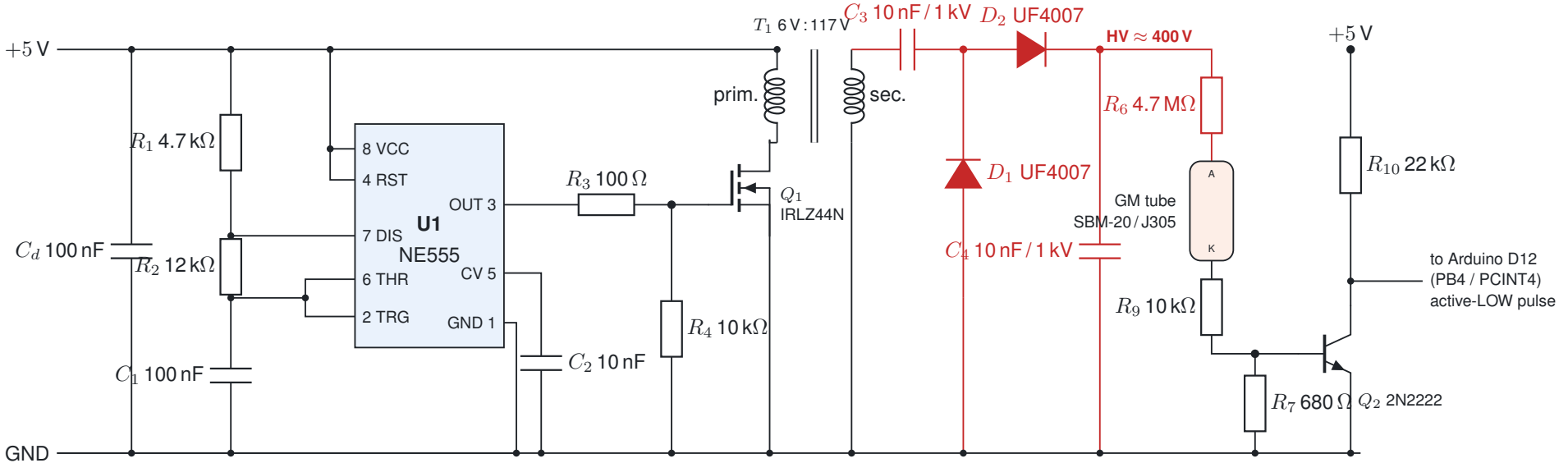


Corrected Geiger–Müller front-end (topology of the 2022 build, element14 "homebrew Geiger counter").

Changes vs. thesis Fig. 8: D_1, D_2 were labelled 2N2222 (a transistor) → UF4007 fast HV diodes; Q_1 IRF220 → logic-level IRLZ44N (fully on at $V_{GS} = 5$ V); gate resistor 100 kΩ →100 Ω with 10 kΩ pull-down; "4.7m" → 4.7 MΩ anode resistor; R_7 650 Ω → 680 Ω (E12); output now taken from the **collector** of Q_2 (the 2022 drawing tapped the base node, ≈0.6 V swing).

555 astable: $f = \dfrac{1.44}{(R_1 + 2R_2)C_1} \approx 0.50$ kHz, duty $= \dfrac{R_1 + R_2}{R_1 + 2R_2} \approx 58\,\%$. Adjust f / duty to reach the tube plateau voltage (SBM-20: 350–475 V); measure with a ≥1 GΩ divider.

DANGER – red nets carry several hundred volts. Capacitors stay charged after power-off. Rate C_3, $C_4 \geq$ 1 kV.